

- ① (a) Uncountable (0.5)
 (b) $\mathbb{Q}^2$ countable (infinite) (0.5)
 (c) $[0, 1]^2$ uncountable (0.5).

② (a)
$$\frac{\binom{4}{c_2}}{\binom{52}{c_2}} \quad (1)$$

(b)
$$\frac{\binom{13}{c_1} \binom{4}{c_2}}{\binom{52}{c_2}} \quad (1)$$

- ③ (a) $P(A \cap B), P(A \cup B)$. can't assume that $A = \Omega$ or $B = \Omega$.
to show this

$$1 = P(\Omega) \geq P(A \cup B) \geq P(A) = 1.$$

0.5

$$\Rightarrow 1 \geq P(A \cup B) \geq 1 \Rightarrow P(A \cup B) = 1$$

0.5

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 1.$$

Alternate method

$$P(A^c \cup B^c) \leq P(A^c) + P(B^c) = 0.$$

$$\Rightarrow P(A^c \cup B^c) = 0 \quad [\text{Non-negativity}].$$

$$\Rightarrow P((A \cap B)^c) = 0 \Rightarrow P(A \cap B) = 1$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 1.$$

④ (b)
$$P\left(\bigcap_{n=1}^{\infty} A_n\right) = P\left(\left(\bigcup_{n=1}^{\infty} A_n^c\right)^c\right)$$

$$= 1 - P\left(\bigcup_{n=1}^{\infty} A_n^c\right).$$

[Using Union bound]

$$\geq 1 - \sum_{n=1}^{\infty} P(A_n^c) = 1.$$

$$\Rightarrow P\left(\bigcap_{n=1}^{\infty} A_n\right) \geq 1$$

But since $\bigcap_{n=1}^{\infty} A_n \subseteq \Omega$.

$$\Rightarrow P\left(\bigcap_{n=1}^{\infty} A_n\right) \leq P(\Omega) = 1.$$

$$\Rightarrow 1 \leq P\left(\bigcap_{n=1}^{\infty} A_n\right) \leq 1$$

$$\Rightarrow P\left(\bigcap_{n=1}^{\infty} A_n\right) = 1$$

4.

(a) Independence of A, B, C.

$$\Rightarrow P(A \cap B \cap C) = P(A)P(B)P(C)$$

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cap C) = P(A)P(C) \Rightarrow P(A|C) = P(A)$$

$$P(B \cap C) = P(B)P(C) \Rightarrow P(B|C) = P(B)$$

$$P(A \cap B | C) = \frac{P(A \cap B \cap C)}{P(C)} = \frac{P(A)P(B)P(C)}{P(C)}$$

$$= P(A)P(B) = P(A|C)P(B|C).$$

$\Rightarrow$ A, B are independent conditioned over C.

0.5

(b) A, B disjoint implies A, B independent

True

False

5.

A: person has disease

B: test result positive

$$P(A|B) = 0.999.$$

$$P(B|A) = x, \quad P(B^c|A^c) = y$$

$$\left(\begin{array}{l} P(A) = 0.01 \Rightarrow P(A^c) = 0.99 \\ \Rightarrow P(B^c|A) = (1-x) \quad P(B|A^c) = 1-y \end{array} \right.$$

$$0.999 = P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}$$

$$0.999 = \frac{x(0.01)}{x(0.01) + (1-y)(0.99)}$$

$$\Rightarrow 0.999 \cdot x(0.01) + (0.999)(1-y)(0.99) = x(0.01)$$

$$\Rightarrow (0.999)(1-y)(0.99) = x(0.01)(0.001)$$

$$x = 10^5 \times (0.999) \times (0.99)(1-y)$$

(a)

Since $x \leq 1$

$$\Rightarrow 10^5 \times (0.999) \times (0.99)(1-y) \leq 1$$

$$\Rightarrow (1-y) \leq \frac{10^{-5}}{(0.999)(0.99)}$$

this is
enough to get
the full marks

$$\Rightarrow y \geq 1 - \frac{10^{-5}}{0.999 \times 0.99}$$

$$\Rightarrow y \geq 0.999989$$

(b)

Suppose $y = 1 - 10^{-6}$

$$\text{Then } x = 10^5 \times (0.999) \times (0.99) \times 10^{-6}$$

will give the required accuracy

$$\Rightarrow \boxed{\alpha = 0.0989}$$

(This is an extreme case where sensitivity requirement is less as specificity guarantees are very high.)

Alternate soln for Qn 3-b

if you want to use continuity Corollary 2 seen in class.

$$B_n = A_1 \cap A_2 \cap \dots \cap A_n$$

$$B_1 \subset B_2 \subset B_3 \dots$$

$$P\left(\bigcap_{n=1}^{\infty} B_n\right) = \lim_{n \rightarrow \infty} P(B_n)$$

$$P(B_n) = P(A_1 \cap A_2 \cap \dots \cap A_n)$$

$$= 1 - P(A_1^c \cup A_2^c \cup \dots \cup A_n^c)$$

$$\geq 1 - \sum_{i=1}^n P(A_i^c)$$

$$= 1$$

$$\Rightarrow P(B_n) \geq 1 \Rightarrow P(B_n) = 1$$

$$\Rightarrow P\left(\bigcap_{n=1}^{\infty} B_n\right) = \lim_{n \rightarrow \infty} P(B_n) = 1$$